Midterm Examination 2: Machine Learning Theory – 2026-I

Professor: Diego Armando Pérez Rosero, Ph.D. Candidate
Department of Electrical, Electronic, and Computer Engineering
National University of Colombia – Manizales Campus

Instructions

- In order to receive full credit, all responses must be clearly justified and must present the corresponding procedures and reasoning step by step, in a concrete, clear, and complete manner.
- The midterm examination must be submitted no later than June 10, 2026, at 18:00. The simulation components must be developed in Python notebooks executed in either the Google Colab or Kaggle environment.
- All code must be properly commented in the relevant code cells and explained in accompanying text cells written in Markdown. Code submissions that do not include comments or discussion will not be considered in the final assessment.
- The midterm examination may be submitted in pairs; however, the grade will be assigned individually.

Questions

1. Prepare an infographic in which you analyze and compare the principal similarities and differences among dense, convolutional, recurrent, and attention-based architectures (Transformers) used in deep learning. The infographic must include, at a minimum: (i) the fundamental mathematical model of each architecture, (ii) an illustrative diagram of its principal layers, and (iii) its most representative applications.

2. Using the dataset available at `https://www.kaggle.com/competitions/amia-public-challenge-2026`, propose a deep learning-based solution. Include a description of the input and output variables, the main objective of the competition, a basic exploratory analysis, and the procedures used to encode missing values and categorical variables. Justify the selected architecture in accordance with the discussion presented in Question 1. Submit a screenshot demonstrating the submission of the best solution identified for the Kaggle competition.

3. Manually implement, from first principles, a neural network for a regression problem with two input attributes, two hidden layers with two neurons each, and one output neuron. Use the ReLU activation function in all neurons, including the output neuron. Explicitly implement the forward propagation, the error computation, the gradient of each parameter, and the update of each weight according to the rule

$$w_{\text{new}} = w_{\text{old}} + \lambda \nabla_w,$$

applying this procedure to all weights and biases of the network. Use the following data simulation procedure: generate $n = 3000$ independent observations with $x_1, x_2 \sim \mathcal{U}(-2, 2)$ and noise $\varepsilon \sim \mathcal{N}(0, 0.15^2)$. Define the true signal as

$$f(x_1, x_2) = \max\{0,\ 1.5 + 2\sin(\pi x_1) + 0.75x_2^2 - 1.2x_1x_2\},$$

and define the observed target variable as

$$y = \max\{0,\ f(x_1, x_2) + \varepsilon\}.$$

Compare the fitted model with both the true signal and the observed values using regression metrics and prediction-versus-observed-value plots. **Extra credit:** compare these results with those obtained using a dense neural network implemented with a high-level library or, alternatively, with a recurrent neural network that treats (x_1, x_2) as a short sequence. Discuss the possible differences in performance, training stability, number of parameters, computational time, capacity to approximate the simulated nonlinearity, and interpretability of the results.